

Technical Note: Data Inversion Theory for the DMA-APM-CPC Tandem Aerosol Measurement System

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This document provides a self-contained description of the mathematical theory and numerical algorithms implemented in the DMA-APM mass distribution inversion tool. It covers the formulation of the forward problem (1D and 2D integral models), the physical justification for the 1D approximation, the original RK4-based APM transfer function simulation, the Chahine-Twomey inversion algorithm with Markowski smoothing, and the Poisson-statistics-based convergence criterion.

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1. Objective and problem statement

The objective of this analysis is to estimate the highly resolved mass distribution of aerosol particles that have been pre-classified by a specific electrical mobility diameter using a DMA-APM-CPC tandem measurement system. Specifically, we seek the mass distribution function at the DMA outlet:

$$f(m) = \frac{dN}{dm}, \quad (1)$$

where N is the particle number concentration and m is the particle mass. The observed CPC concentration $n(V)$ at each APM applied voltage V is related to $f(m)$ through a Fredholm integral equation of the first kind, and recovering $f(m)$ from $n(V)$ constitutes an ill-posed inverse problem.

2. Formulation of the forward problem

2.1 The rigorous 2D-integral model

The expected particle number concentration $n(V)$ observed downstream of the APM is governed by a double integral over both mass m and electrical mobility Z_p :

$$n(V) = \int_0^\infty \int_0^\infty \Omega_{\text{APM}}(m, V, Z_p) f_{\text{in}}(m, Z_p) \Omega_{\text{DMA}}(Z_p) dZ_p dm, \quad (2)$$

where $f_{\text{in}}(m, Z_p)$ is the intrinsic 2D mass-mobility distribution of the raw aerosol entering the DMA, $\Omega_{\text{DMA}}(Z_p)$ is the DMA transfer function, and $\Omega_{\text{APM}}(m, V, Z_p)$ is the APM transfer function.

Assuming that f_{in} is approximately constant with respect to Z_p over the narrow DMA transmission window, we define the target distribution $f(m)$ as the mass distribution of the aerosol exiting the DMA:

$$f(m) = f_{\text{in}}(m, Z_p^*) \int_0^\infty \Omega_{\text{DMA}}(Z_p) dZ_p. \quad (3)$$

Using this definition, the forward problem reduces to:

$$n(V) = \int_0^\infty K_{\text{eff}}^{2D}(V, m) f(m) dm, \quad (4)$$

where the effective 2D kernel is:

$$K_{\text{eff}}^{2D}(V, m) = \frac{\int_0^\infty \Omega_{\text{APM}}(m, V, Z_p) \Omega_{\text{DMA}}(Z_p) dZ_p}{\int_0^\infty \Omega_{\text{DMA}}(Z_p) dZ_p}. \quad (5)$$

In the implementation, the DMA transfer function is modelled as a triangular function centered at the target mobility Z_p^* with half-width βZ_p^* , where $\beta = Q_{\text{sample}}/Q_{\text{sheath}}$ is the DMA flow ratio. The integral in Eq. (5) is evaluated numerically using the rectangle rule with a configurable number of quadrature points.

2.2 The 1D-integral approximation

By approximating the DMA output as a Dirac delta function centered at the target mobility Z_p^* , the effective kernel simplifies to:

$$K^{1D}(V, m) = \Omega_{\text{APM}}(m, V, Z_p^*), \quad (6)$$

and the forward problem becomes:

$$n(V) \approx \int_0^\infty K^{1D}(V, m) f(m) dm. \quad (7)$$

In both the 1D and 2D models, the inversion algorithm solves for the same target: $f(m)$, the mass distribution of the particles at the DMA outlet.

3. Physical justification: why the 1D model is practically sufficient

In practice, the mass distributions obtained from the 1D and 2D models are virtually identical, even when the DMA resolution is relatively broad (e.g., $\beta = 0.2$). This equivalence arises from the fundamental classification principle of the APM. The central classified mass is:

$$m_c = \frac{eV}{r_c^2 \omega^2 \ln(r_2/r_1)}, \quad (8)$$

where e is the elementary charge, $r_c = (r_1 + r_2)/2$ is the gap centre radius, ω is the angular velocity, and r_1, r_2 are the inner and outer cylinder radii. This expression is completely independent of the

particle's electrical mobility Z_p . A variation in Z_p (due to the DMA's finite transmission width) only alters the particle's radial drift velocity inside the APM, slightly affecting its transit time. Consequently, Z_p variations marginally broaden the width of Ω_{APM} but do not shift its central peak. Because the 2D convolution averages these minor, symmetric width changes, the resulting K_{eff}^{2D} is nearly indistinguishable from K^{1D} .

The 1D approximation therefore provides mathematically and physically robust results at approximately $1/N_{Z_p}$ of the computational cost of the 2D model, where N_{Z_p} is the number of Z_p quadrature points.

4. APM transfer function: particle trajectory simulation

4.1 Equation of motion

To compute $\Omega_{\text{APM}}(m, V, Z_p)$, this tool employs an original numerical simulation method that directly tracks particle trajectories within the APM gap using the 4th-order Runge-Kutta (RK4) method.

Inside the APM annular gap (inner radius r_1 , outer radius r_2 , centre radius $r_c = (r_1 + r_2)/2$, half-width $\delta = (r_2 - r_1)/2$), a particle of mass m and electrical mobility Z_p experiences centrifugal and electrostatic forces. Under the assumption of a fully developed parabolic laminar flow profile, the ratio of the radial to axial velocity components is (Ehara et al., 1996):

$$\frac{dr}{dz} = \frac{v_r(r)}{v_z(r)} = \frac{\frac{Z_p}{e} \left(m\omega^2 r - \frac{eV}{r \ln(r_2/r_1)} \right)}{\frac{3Q}{4\pi\delta r_c} \left[1 - \left(\frac{r-r_c}{\delta} \right)^2 \right]}, \quad (9)$$

where Q is the volumetric aerosol flow rate. This can be rearranged to:

$$\frac{dr}{dz} = \frac{8\pi\delta r_c}{3Q} \frac{Z_p}{e} \cdot \frac{m\omega^2 r - \frac{eV}{r \ln(r_2/r_1)}}{1 - \left(\frac{r-r_c}{\delta} \right)^2}. \quad (10)$$

4.2 RK4 integration and transmission efficiency

The trajectory equation (Eq. 10) is integrated along the axial coordinate z from $z = 0$ to $z = L$ (the electrode length) using the classical 4th-order Runge-Kutta method with a fixed step size Δz :

$$k_1 = \Delta z \cdot g(r_n), \quad (11a)$$

$$k_2 = \Delta z \cdot g(r_n + k_1/2), \quad (11b)$$

$$k_3 = \Delta z \cdot g(r_n + k_2/2), \quad (11c)$$

$$k_4 = \Delta z \cdot g(r_n + k_3), \quad (11d)$$

$$r_{n+1} = r_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4), \quad (11e)$$

where $g(r) \equiv dr/dz$ as defined in Eq. (10).

A total of N_{r_0} particles are launched from uniformly spaced initial radial positions $r_0 \in [r_1, r_2]$. A particle is considered lost if it hits the inner or outer electrode wall at any step. The transmission efficiency is computed as the flux-weighted fraction of surviving particles:

$$\Omega_{\text{APM}} = \frac{\sum_{r_0 \in \text{survived}} w(r_0)}{\sum_{r_0 \in \text{all}} w(r_0)}, \quad (12)$$

where the weight $w(r_0)$ accounts for the parabolic flow profile:

$$w(r_0) = \frac{3}{2} \left[1 - \left(\frac{r_0 - r_c}{\delta} \right)^2 \right]. \quad (13)$$

4.3 Cunningham slip correction

The electrical mobility Z_p is related to the mobility diameter D_{mob} via:

$$Z_p = \frac{e C_c(D_{\text{mob}})}{3\pi\eta D_{\text{mob}}}, \quad (14)$$

where η is the dynamic viscosity of air and C_c is the Cunningham slip correction factor (Hinds, 1999):

$$C_c = 1 + \frac{1}{Pd} [15.60 + 7.00 \exp(-0.059 Pd)], \quad (15)$$

with P the atmospheric pressure in kPa and d the diameter in μm .

4.4 Kernel matrix discretisation

For the discrete inversion, the integral in Eq. (7) is approximated by the rectangle rule over J mass bins:

$$n(V_i) \approx \sum_{j=1}^J K_{i,j} f(m_j), \quad (16)$$

where $K_{i,j} = \Omega_{\text{APM}}(m_j, V_i, Z_p^*) \cdot \Delta m$ for the 1D model, m_j are uniformly spaced mass grid points, and $\Delta m = (m_{\text{max}} - m_{\text{min}})/(J - 1)$. The kernel matrix $\mathbf{K}$ has shape $I \times J$, where I is the number of voltage bins.

5. Inversion algorithm: Chahine-Twomey method with internal smoothing

5.1 Update equation

To solve the discrete ill-posed problem $\mathbf{n} = \mathbf{K}\mathbf{f}$, we use the non-negative Chahine-Twomey iterative algorithm (Twomey, 1975). The update rule at iteration k is:

$$f_j^{(k+1)} = f_j^{(k)} \cdot \frac{\sum_{i=1}^I \hat{K}_{i,j} (n_{\text{meas},i} / n_{\text{calc},i}^{(k)})}{\sum_{i=1}^I \hat{K}_{i,j}}, \quad (17)$$

where $\hat{K}_{i,j} = K_{i,j} / \max(\mathbf{K})$ is the normalised kernel, $n_{\text{meas},i}$ is the measured concentration in the i -th voltage bin, and $n_{\text{calc},i}^{(k)} = \sum_j K_{i,j} f_j^{(k)}$ is the forward-calculated concentration at iteration k .

The initial guess is a uniform distribution:

$$f_j^{(0)} = \frac{\sum_{i=1}^I n_{\text{meas},i}}{m_{\text{max}} - m_{\text{min}}}, \quad j = 1, \dots, J. \quad (18)$$

5.2 Markowski 1-2-1 smoothing

After each iteration, a three-point moving average filter (Markowski, 1987) is applied to suppress high-frequency oscillations caused by noise amplification:

$$\tilde{f}_j = \frac{1}{4}f_{j-1} + \frac{1}{2}f_j + \frac{1}{4}f_{j+1}, \quad j = 2, \dots, J-1. \quad (19)$$

The endpoints ($j = 1$ and $j = J$) are not smoothed.

6. Poisson variance and convergence criterion

6.1 Poisson counting statistics

The CPC measures particle counts N_i during the integration time $t_{\text{meas},i}$ at each voltage bin. The measured concentration is $n_{\text{meas},i} = N_i / V_{\text{sample},i}$, where $V_{\text{sample},i} = Q_{\text{CPC}} \cdot t_{\text{meas},i}$ is the total sampled air volume. Because particle counting follows Poisson statistics, the variance of the measured concentration is:

$$\text{Var}(n_i) = \frac{n_{\text{calc},i}}{V_{\text{sample},i}}. \quad (20)$$

In practice, a floor is applied to prevent division by zero:

$$\text{Var}(n_i) = \max\left(\frac{n_{\text{calc},i}}{V_{\text{sample},i}}, \frac{1}{V_{\text{sample},i}^2}\right). \quad (21)$$

6.2 Reduced chi-squared criterion

Convergence is determined by the reduced chi-squared statistic:

$$\chi^2 = \frac{1}{I} \sum_{i=1}^I \frac{(n_{\text{meas},i} - n_{\text{calc},i})^2}{\text{Var}(n_i)}. \quad (22)$$

The iteration terminates when $\chi^2 < \chi_{\text{threshold}}^2$. The default threshold is $\chi_{\text{threshold}}^2 = 1.0$, corresponding to the expectation that residuals should be comparable to the measurement noise. This criterion

prevents overfitting: stopping at $\chi^2 \approx 1$ ensures that the solution explains the data to within the precision allowed by Poisson counting noise.

7. Data preprocessing: voltage binning

7.1 Merging upward and downward scans

The raw APM data consists of time-series measurements during continuous voltage scanning (both upward and downward sweeps). The data are binned into I voltage bins by grouping measurements with similar applied voltages.

7.2 Binning rules for physical quantities

The binning procedure strictly distinguishes between intensive and extensive variables:

- **Intensive variables** (state quantities): Applied voltage V and particle concentration n are averaged within each bin using the arithmetic mean.
- **Extensive variables** (amount quantities): The measurement time t_{meas} for each bin is obtained by summing the individual time intervals Δt of each data row assigned to that bin.

This distinction is critical for the correct evaluation of $V_{\text{sample},i} = Q_{\text{CPC}} \cdot t_{\text{meas},i}$, which directly enters the Poisson variance calculation (Eq. 21).

8. Implementation mapping

Theory	Implementation	File
Cunningham correction, Eq. (15)	<code>_cunningham(Dmob)</code>	<code>kernel_simulator.py</code>
Trajectory equation, Eq. (10)	<code>dr_dz(rad)</code> inside <code>_rk4_transmission</code>	<code>kernel_simulator.py</code>
RK4 integration, Eq. (11)	<code>_rk4_transmission(...)</code>	<code>kernel_simulator.py</code>
Transmission efficiency, Eq. (12)-(13)	<code>_rk4_transmission(...)</code>	<code>kernel_simulator.py</code>
1D kernel matrix, Eq. (16)	<code>build_kernel_1d(data, params)</code>	<code>kernel_simulator.py</code>
2D effective kernel, Eq. (5)	<code>build_kernel_2d(data, params)</code>	<code>kernel_simulator.py</code>
Chahine-Twomey update, Eq. (17)	<code>solve_chahine_twomey(...)</code>	<code>inversion_solver.py</code>
Markowski smoothing, Eq. (19)	<code>solve_chahine_twomey(...)</code>	<code>inversion_solver.py</code>
Poisson variance, Eq. (21)	<code>solve_chahine_twomey(...)</code>	<code>inversion_solver.py</code>
Chi-squared criterion, Eq. (22)	<code>solve_chahine_twomey(...)</code>	<code>inversion_solver.py</code>
Voltage binning, Sec. 7	<code>load_and_bin(params)</code>	<code>data_parser.py</code>
Gaussian mode fitting	<code>fit_gaussian_mode(m_array, f)</code>	<code>visualization.py</code>

9. References

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